VISHWAKARMA INSTITUTES

CODE APEX 3.0

AISF
ARTIFICIAL INTELLIGENCE STUDENT FORUM

# TREVYN

## India's Autonomous QA Engineer

**Discover. Verify. Repair. Deliver with confidence.**

iMocha

Organised by AISF · Vishwakarma Institute of Technology, Pune

# THE CONNECTED DOTS

## A TYPICAL MULTI-TOOL QA WORKFLOW

e.g. duplicate orders after a checkout change

## BEFORE VS WITH TREVYN

## HUMANS STILL DECIDE

- High-impact tests
- Uncertain diagnoses
- Accepting patches
- Release and risk calls

**92%+**
use AI in functional testing
up from 60% a year earlier

**29%**
report more or more severe defects
same survey

**86%**
rate human involvement as vital
same survey

**Source:** Applause, 2026 State of Digital Quality in Functional Testing, 30 Sep 2026 [2]. Survey findings, not universal measurements.

# THE MISSING LAYER

ISOLATED TEST RESULTS

Logs · screenshots · failures
from separate tools

1 **Understand**
URL, repo, requirements

2 **Plan**
Valid, executable tests

3 **Execute**
The right tool, every time

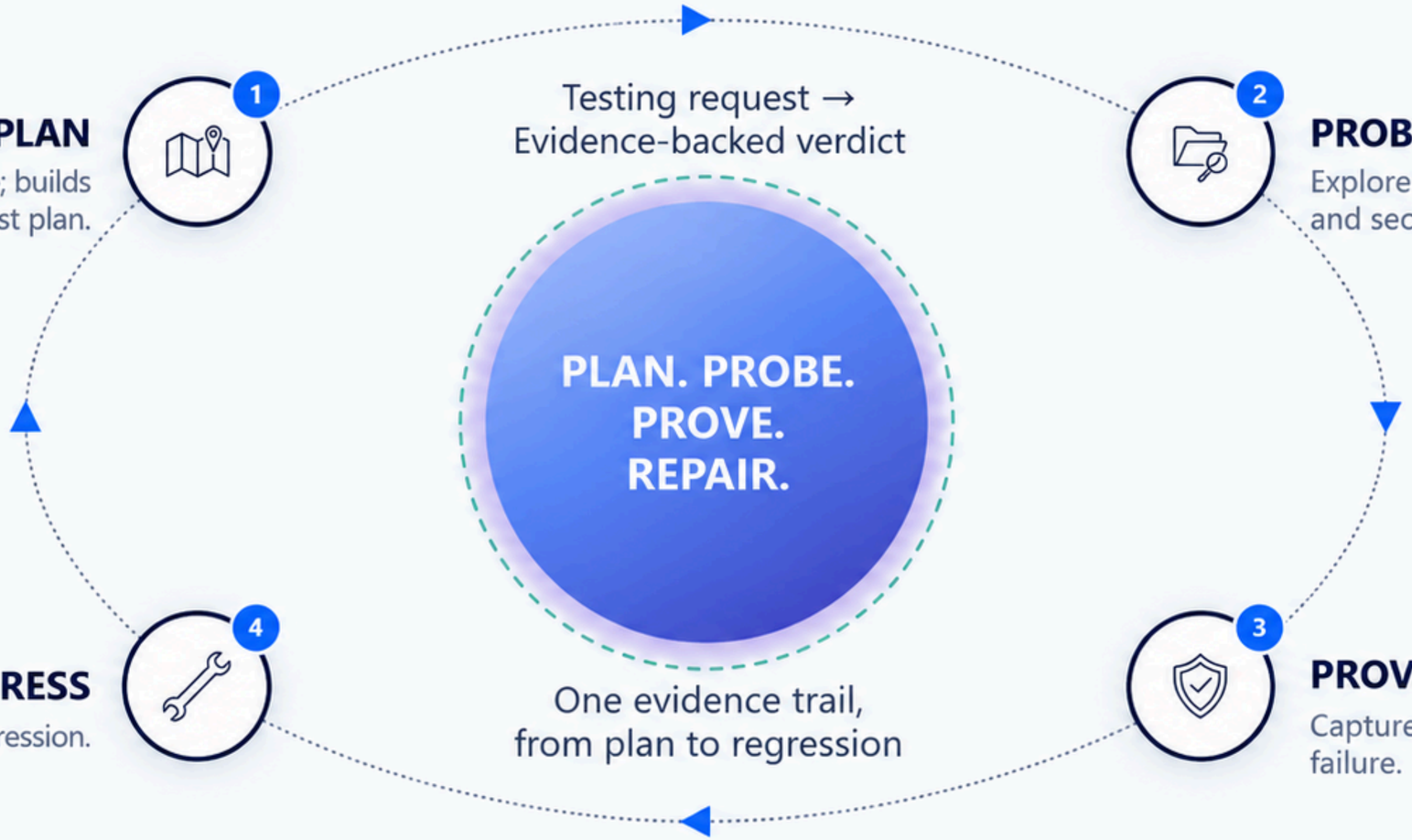
4 **Verify**
Results checked against evidence

5 **Resolve**
Repair proposed, then rerun

## AI proposes. Evidence verifies. Humans decide.

Trevyn coordinates Playwright, k6 and other mature tools. It does not replace them.

# MEET TREVYN

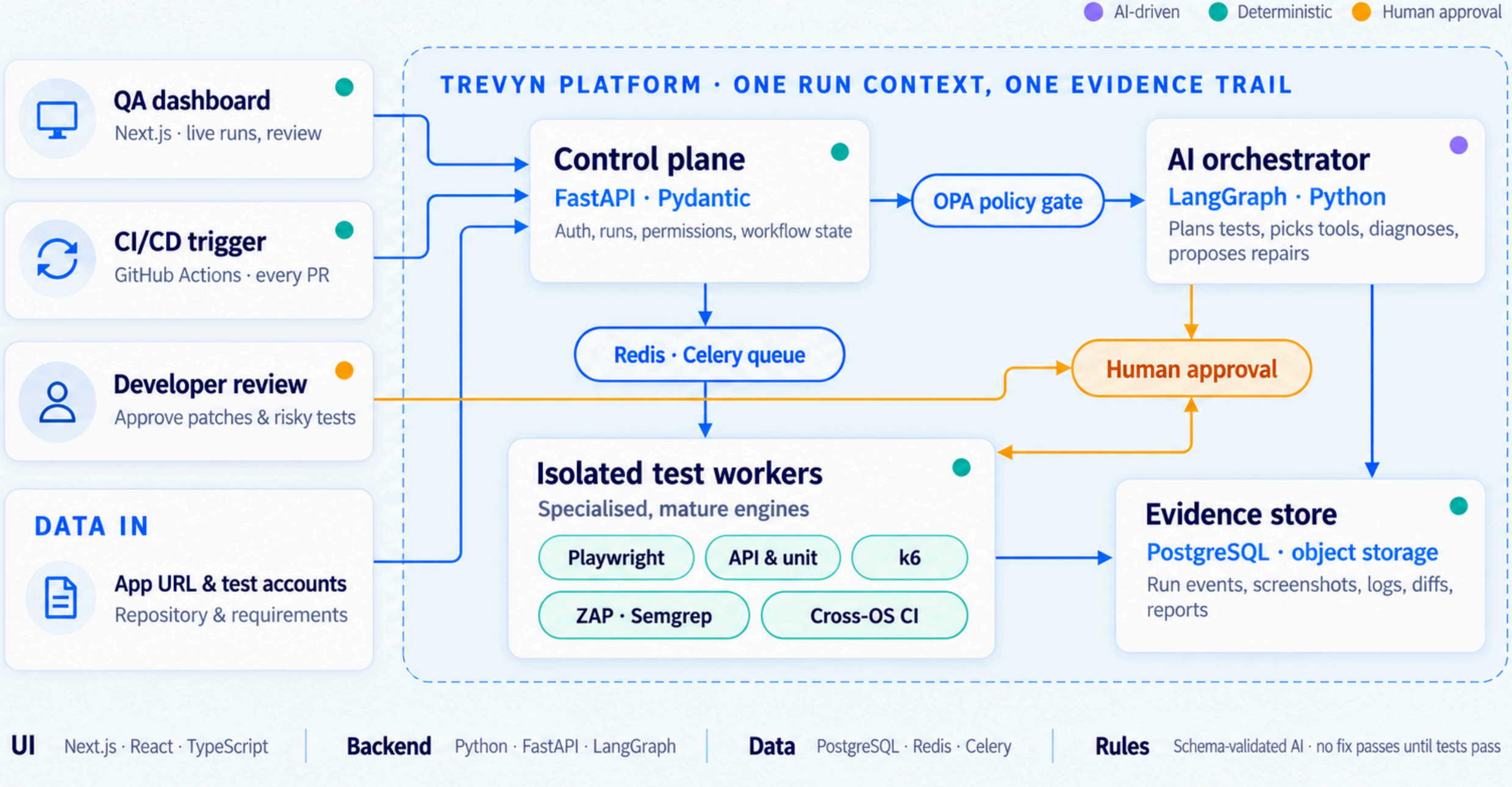



PRIMARY OUTPUTS

- Reusable automated tests
- Evidence-backed bug reports
- Pass / fail / inconclusive
- Repair suggestions
- Regression run history

Built around the IM-02 brief: explore, decide, execute, report with evidence, reuse tests [1].

# SYSTEM ARCHITECTURE



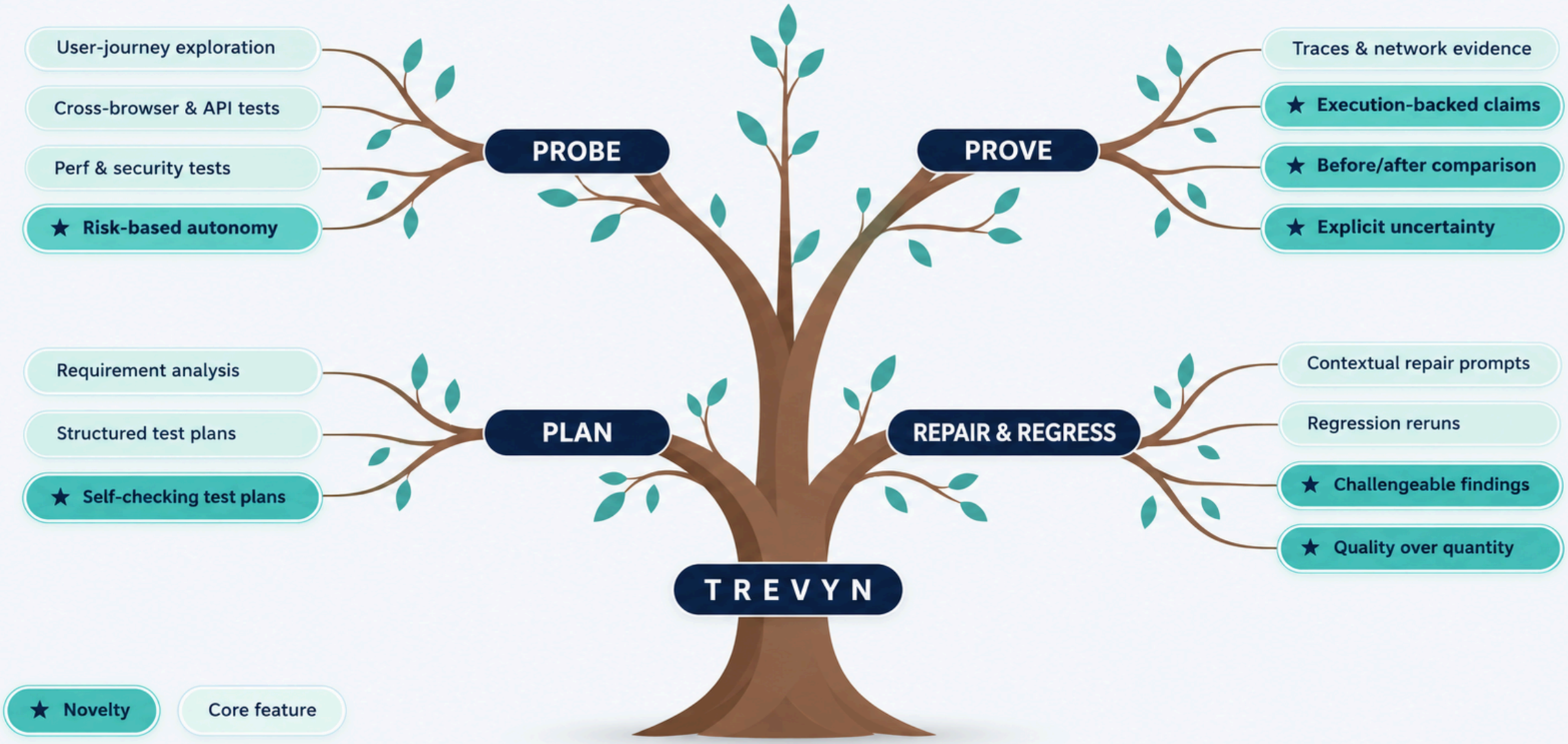

**UI** Next.js · React · TypeScript | **Backend** Python · FastAPI · LangGraph | **Data** PostgreSQL · Redis · Celery | **Rules** Schema-validated AI · no fix passes until tests pass

# TECHNOLOGY STACK

| | LAYER | TOOLS | WHY |
|---|---|---|---|
| 01 | **Frontend** | Next.js · React · TypeScript · Tailwind · shadcn/ui · React Flow | Dashboard and live runs |
| 02 | **Backend & AI** | Python · FastAPI · Pydantic · LangGraph · JSON Schema | Structured orchestration |
| 03 | **Test engines** | Playwright · k6 · OWASP ZAP · Semgrep · pytest · Vitest · JUnit | Mature tools, not reinvented |
| 04 | **Data & jobs** | PostgreSQL · Redis · Celery · Object storage · Docker | History, queues, evidence |
| 05 | **Security & CI** | OPA / Rego · RBAC · Managed secrets · GitHub Actions | Policy and automation |
| 06 | **AI models** | OpenAI API · Anthropic API · provider-agnostic | Quality, speed, cost |

Open-source-first. Hosted services and model APIs may add costs; ChatGPT Plus and Claude Pro do not include API credits.

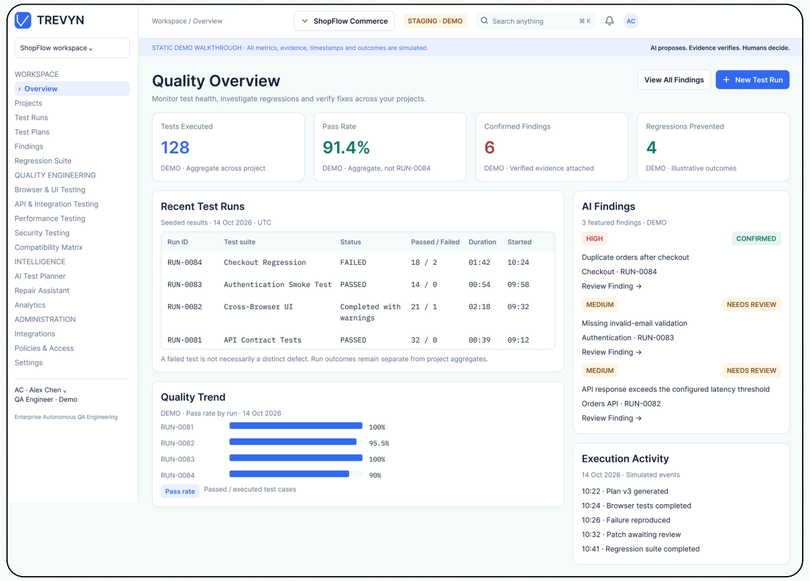
# FEATURES & NOVELTIES

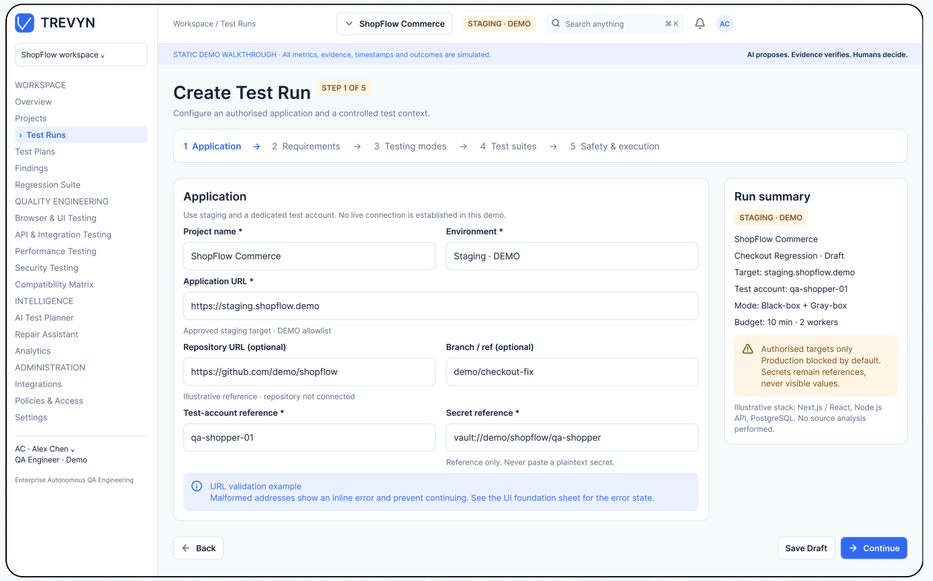


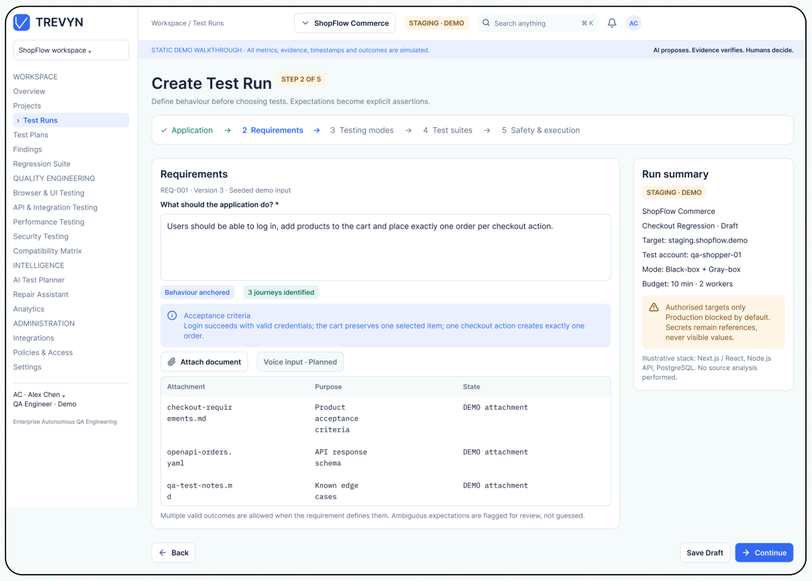

★ Novelty

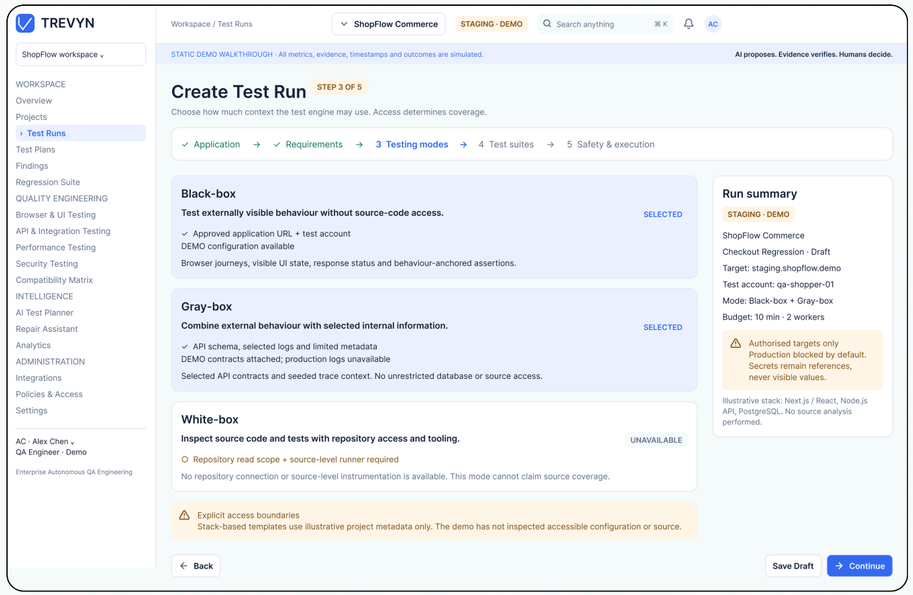
Core feature

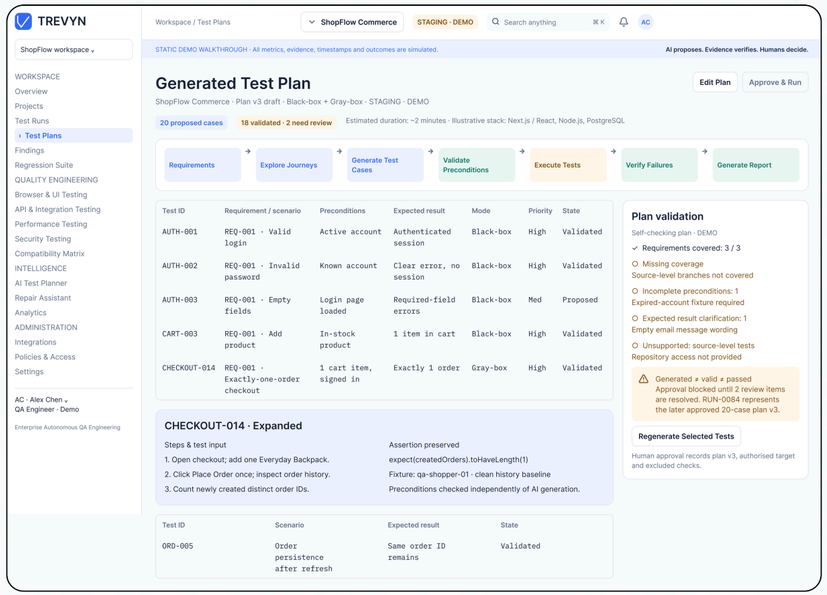
# PROTOTYPE WALKTHROUGH

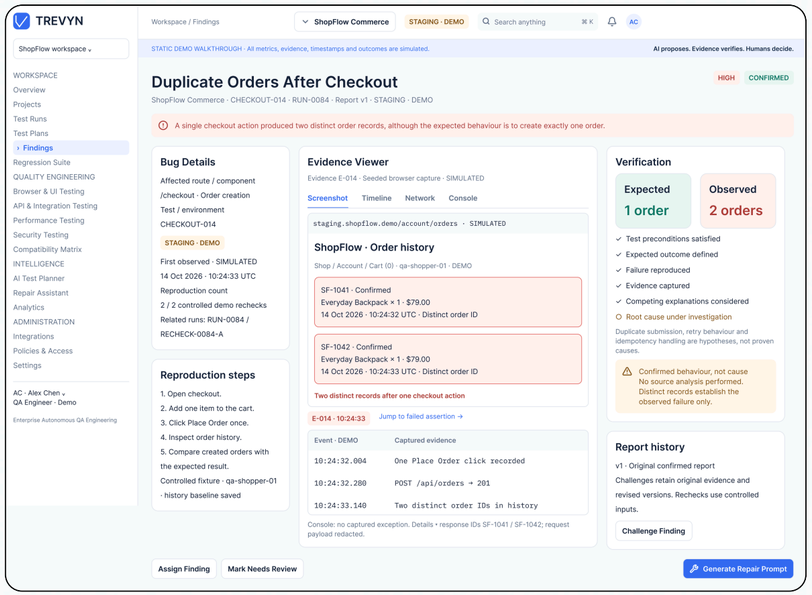



[Prototype]

# ENTERPRISE READINESS

1 → 2 → 3 → 4

## 1 PROTOTYPE

**Q4 2026**

- **Docker-isolated** test workers
- **Playwright** traces + regression
- **Synthetic data**, seeded defects

**EXIT GATE**
Reproducible end-to-end reports

## 2 TEAM PILOT

**Q1 2027**

- **RBAC** per project
- **OPA / Rego** target allowlists
- **Human-in-the-loop** patch review

**EXIT GATE**
Safety measured on seeded defects

## 3 ENTERPRISE PILOT

**Q2 – Q3 2027**

- **SSO / OIDC**, tenant isolation
- **Secrets** + egress control
- **CI/CD gates**, usage budgets

**EXIT GATE**
Security and ops reviews passed

## 4 PRODUCTION

**Q4 2027+**

- **Pen testing**, security review
- **SLOs**, backup + DR drills
- **Audit logs**, governance gates

**EXIT GATE**
Approved for production use

**READINESS** | Security · Governance · Observability · CI/CD · Recovery · Cost controls | Proposed timeline · NIST SSDF [6]

# REFERENCES


1 **Official problem statement**
CODE APEX 3.0, iMocha Track, IM-02: Autonomous AI QA Engineer, p. 5; guidelines, p. 9.
unstop.com/competitions/1751374

2 **Industry evidence**
Applause, 2026 State of Digital Quality in Functional Testing, 30 Sep 2026.
applause.com/press-release/2026-state-of-digital-quality-functional-testing

3 **Browser testing & diagnostics**
Playwright docs: Projects, cross-browser setup, Trace Viewer.
https://playwright.dev/docs/trace-viewer

4 **Performance testing**
Grafana k6 docs: Thresholds.
grafana.com/docs/k6/latest/using-k6/thresholds

5 **Web application security**
OWASP Web Security Testing Guide.
owasp.org/projects/web-security-testing-guide

6 **Secure software development**
NIST SP 800-218, SSDF v1.1.
csrc.nist.gov/pubs/sp/800/218/final

7 **Policy-as-code**
Open Policy Agent docs: Rego policy language.
openpolicyagent.org/docs/policy-language

# THANK YOU.

TEAM

## DIMAAG KA DAHI

Vishwakarma Institute of Technology, Pune

MEMBERS

**Rohit Kadam**

**Harsh Dhavane**

**Abhishek Patil**

**Aditya Kachwaha**

Prototype